

Summary

$$y = f(\theta, x) + \epsilon$$

y = measured print response

θ = (Voltage, F-r, dt2, Coverage?)

x = spatial position in the printer

ϵ = noise

with:

$$\theta = (V, F-r, dt2, c?)$$

and:

$$x = (h, n)$$

h = head position $\in \{1, 2, \dots, 120\}$

n = nozzle $\in \{1, 2, \dots, 33600\}$

My questions

$x = (h, n) \rightarrow$ am I implying a discrete product of independent axis?

$\rightarrow$ would it be more accurate to think of it as a physical plane, a continuous space surface?

$\rightarrow$ if we introduce the effect from neighbouring nozzles, would it be accurate to introduce an interaction component? such as:

$$W_{ij}(\theta)$$

where W_{ij} = coupling weight that answers how much does nozzle j influence nozzle i . Since it's probably not fixed, (θ) means the coupling changes with operating conditions. So the sum of the neighbouring nozzles could be

$$\sum_j^{\text{fixed}} W_{ij} s_{\theta}(x_j)$$

$$\text{or } \sum_j^{\text{variable}} W_{ij}(\theta) s_{\theta}(x_j)$$

~~which would result in:~~

with $s(x)$ being the output of SD values as a vector.

$$\cancel{y(x_i) = s}$$

to summarise:

$$y(x_i) = s(x_i) + \sum_j w_{ij}(x_j) + \epsilon_i$$

or

$$y(x_i) = s(x_i) + \sum_j w_{ij}(\theta)(x_j) + \epsilon_i$$

and

$$y(x_i) = f(\theta, x_i)$$

so:

$$f(\theta, x) = s(x_i) + \sum_j w_{ij}(\theta)(x_j) + \epsilon_i$$